SAMI Assignment:
Interaction Audit & Redesign Challenge

Scott Schmidl
sschmidl3@gatech.edu

1 THE INTERACTION AUDIT

1.1 Matchmaking recommendations

When I opted in with #connectme on the Introduce Yourself thread, SAMI replied with a list of classmates it recommended I connect with. In my case, it suggested

- Xi Ling — shared interests: computer programming, CS 7650: Natural Language Processing, the -05:00 timezone.
- Yitao Chen — shared interests: CS 7637: Knowledge-Based Artificial Intelligence, the -05:00 timezone, living in or near Texas.
- Hoyeol Kim — shared interests: CS 7650: Natural Language Processing, the -05:00 timezone, living in or near Texas.
- Johannes Habel — shared interests: CS 7637: Knowledge-Based Artificial Intelligence, CS 7650: Natural Language Processing, the -05:00 timezone.
- Paing Hein Khant — shared interests: living in or near Indiana, CS 7650: Natural Language Processing, living in or near United States.

These recommendations were clearly grounded in the attributes we gave it, while at the same time being somewhat only surface level. SAMI focused on courses, location, and general computer programming interests. It didn't match us based on specific research interests, similar professional backgrounds, or shared future plans. I have not yet had meaningful interaction with these matches beyond seeing their names and shared interests, so I would describe the recommendations as plausible, but not that insightful. I could tell that extra effort would be required, because they didn't on their own reveal connections that would've been hard to find. For example, listing CS7637, a class that we're all currently in, as a common interest.

1.2 Explanations of reasoning (#samiexplain)

I used #samiexplain in a couple of ways to probe SAMI's reasoning. First, I asked it to "explain why and how you matched me with an individual", and it responded by restating key pieces of my introduction:

> I matched you with an individual based on the information you shared in your introduction post, where you mentioned you're Scott Joseph Schmidl II from Texas, a Navy veteran with experience working on nuclear reactors on an aircraft carrier. You're currently a senior software engineer with interests in computer programming and computer vision, and you're pursuing studies in AI-related fields. This background helped identify a suitable match with similar interests and experiences.

It didn't, however, name a specific match or list the precise features that drove the match.

Next, I asked "Hi Sami , What do you know about me so far? #samiexplain". This time SAMI simply quoted my introduction almost verbatim and then described its processing pipeline:

> You said this in your introduction post on the Ed forum: "Hi all. I'm Scott. I live in Texas. I am a Navy veteran. While serving, I worked with nuclear reactors on an aircraft carrier. This is my 11th class in the program (dropped GA last semester and switched to AI spec so I could take KBAI which is more interesting to me). I've completed BS, DO, ML, NLP, DL, CV, CN, SDP, and AIES. Hopefully I'll graduate this semester because I'm jumping right into my MBA at University of Notre Dame in the Fall. I'm currently a senior software engineer at an IT compa. . . ". Here's how SAMI processed it: when you posted, our NLP tool (spaCy) read your post and identified "United States" as a location. SAMI then saved that to its knowledge graph (Neo4j), linked to your User node. When you ask a question, SAMI looks up your User node, follows the relevant relationships, and reports what it finds.

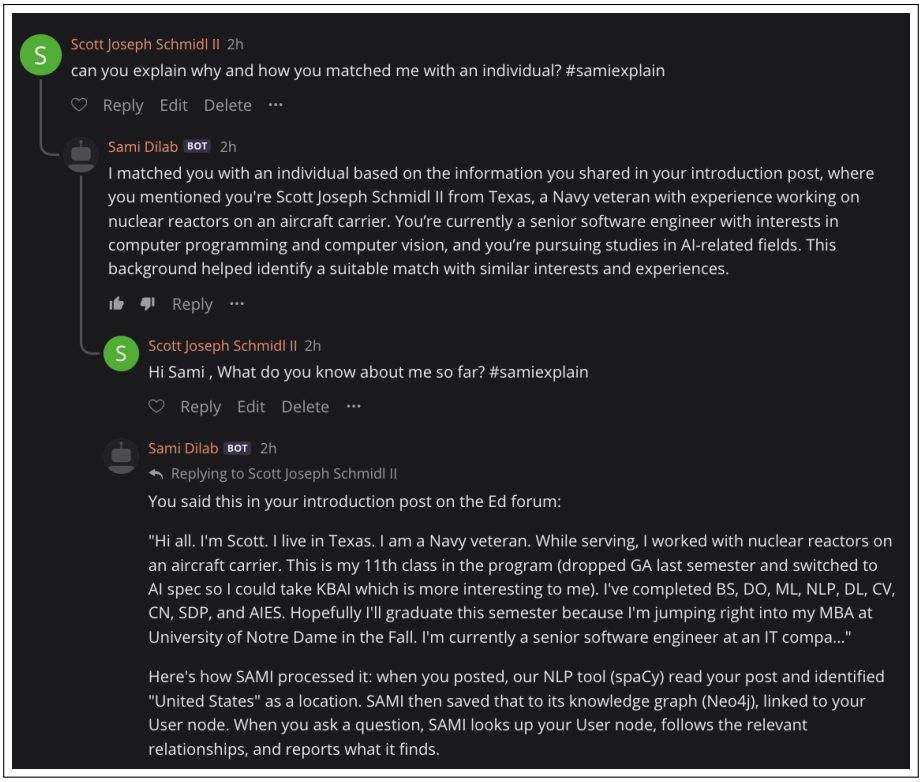


Figure 1—What Does It Know

This gave me a clearer high-level model of how SAMI works, which is essentially an entity extractor feeding a knowledge graph, but it didn't show me the full set of entities it stored about me.

Overall, the explanations were comprehensible and somewhat useful, but not very deep because it just repeated back what I said. SAMI made the existence of spaCy and Neo4j clear, but the matching explanation didn't distinguish between different matches or explain why I was paired with this person over that person with a similar background. At one point it described my specialization as Interactive Intelligence, even though after I corrected it, SAMI claimed my specialization was already Artificial Intelligence and that Interactive Intelligence was not present, which makes it hard to trust.

1.3 SAMI's model of you (#correctme)

To inspect SAMI's model of me, I asked: "Please tell me everything you know about me". It replied with a detailed profile:

> Here's what I know about you, Scott Joseph Schmidl II: You live in the United States with a time zone of -05:00, and you have other locations listed as Notre Dame and Indiana. Your hobbies include computer programming and computer vision. You have taken several courses such as Knowledge-Based Artificial Intelligence, Computer Vision, Natural Language Processing, Machine Learning, Deterministic Optimization, Deep Learning, Software Development Process, Computer Networks, and AI, Ethics, and Society. Your specialization is Interactive Intelligence. Your posts mention your background as a Navy veteran working with nuclear reactors on an aircraft carrier, your academic progress, and your current role as a senior software engineer working with computer vision, specifically OCR and object detection. You also plan to start an MBA at the University of Notre Dame in the Fall.

The model was quite accurate and showed that it's good at extracting more than just a single entity from my post. However, it also contains important inaccuracies, i.e. I don't live at Notre Dame or in Indiana, I live in Texas, and my specialization is Artificial Intelligence not Interactive Intelligence. These errors directly affect the quality of matchmaking and how well my profile represents me to potential connections with classmates.

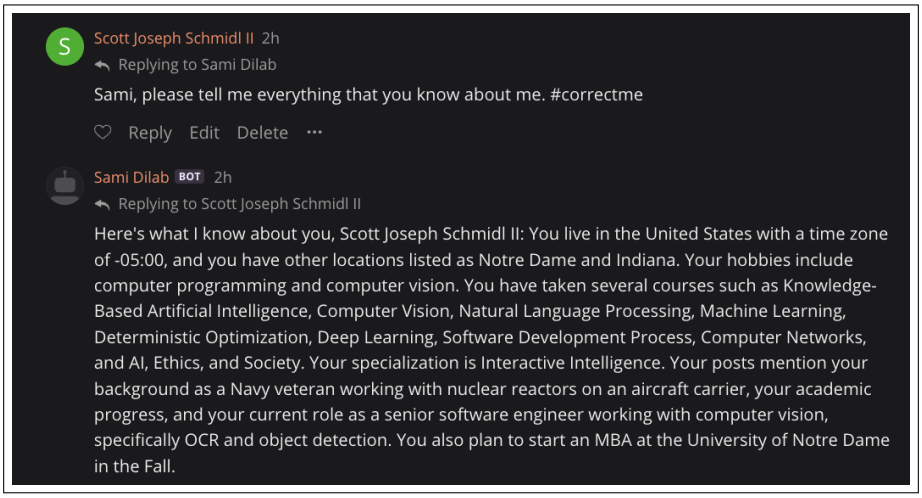


Figure 2—Says Spec is II

1.4 Explanations of revisions

I tried to correct these errors using #correctme and paid attention to how SAMI explained its revisions. I first sent a combined correction: "SAMI, I don't live at Notre Dame or in Indiana. I live in Texas. My specialization isn't Interactive Intelligence, it's Artificial Intelligence. #correctme". SAMI responded with "I tried but couldn't complete that correction within my step budget. Please try simpler instructions, one change at a time". This suggests that its update routine has specific constraints on how much it can process in a single interaction.

Following its recommendation, I first asked for location correction and then specialization. I wrote, "SAMI, I don't live at Notre Dame or in Indiana. I live in Texas. #correctme", but it replied with "Texas is set, I couldn't remove Indiana due to system limitations". For specialization, I wrote "SAMI, my specialization isn't Interactive Intelligence. My specialization is Artificial Intelligence. #correctme", to which it said that my spec was already set to AI and that II wasn't present. This directly contradicted its earlier statement, revealing an inconsistency between its explanations and knowledge graph.

These revisions were readable and somewhat useful, but didn't increase my trust in SAMI. It's good that the primary location is correct, but having Notre Dame and Indiana in "Other" is simply wrong. This can negatively affect matchmaking if it uses this information to connect me with classmates. Furthermore, it's troubling to me that it can't be removed. SAMI, also, didn't show what changed internally after each correction, so I have to trust to some degree rather than verifying that the model matches the truth.

I, then, asked SAMI to explain what changes you made. Instead of telling me what changes it made to my profile, it gave me a long description about setting up SAMI modules saying, "To set up SAMI modules, I first get everything ready so the system can start processing discussion forum posts from your courses. This step is important because it organizes the information needed to connect students based on shared interests and classes. Once the modules are set up successfully, I move on to analyzing those forum posts to find common connections. Since I don't have details beyond this point, I can't explain the full process, but setting up the modules is the crucial first step to get things going.". This seems as though it misunderstood my question word, mistaking "what" for "how" it makes changes and highlights that even when I explicitly ask about changes SAMI falls back to

high-level pipeline descriptions instead of giving me my profile details.

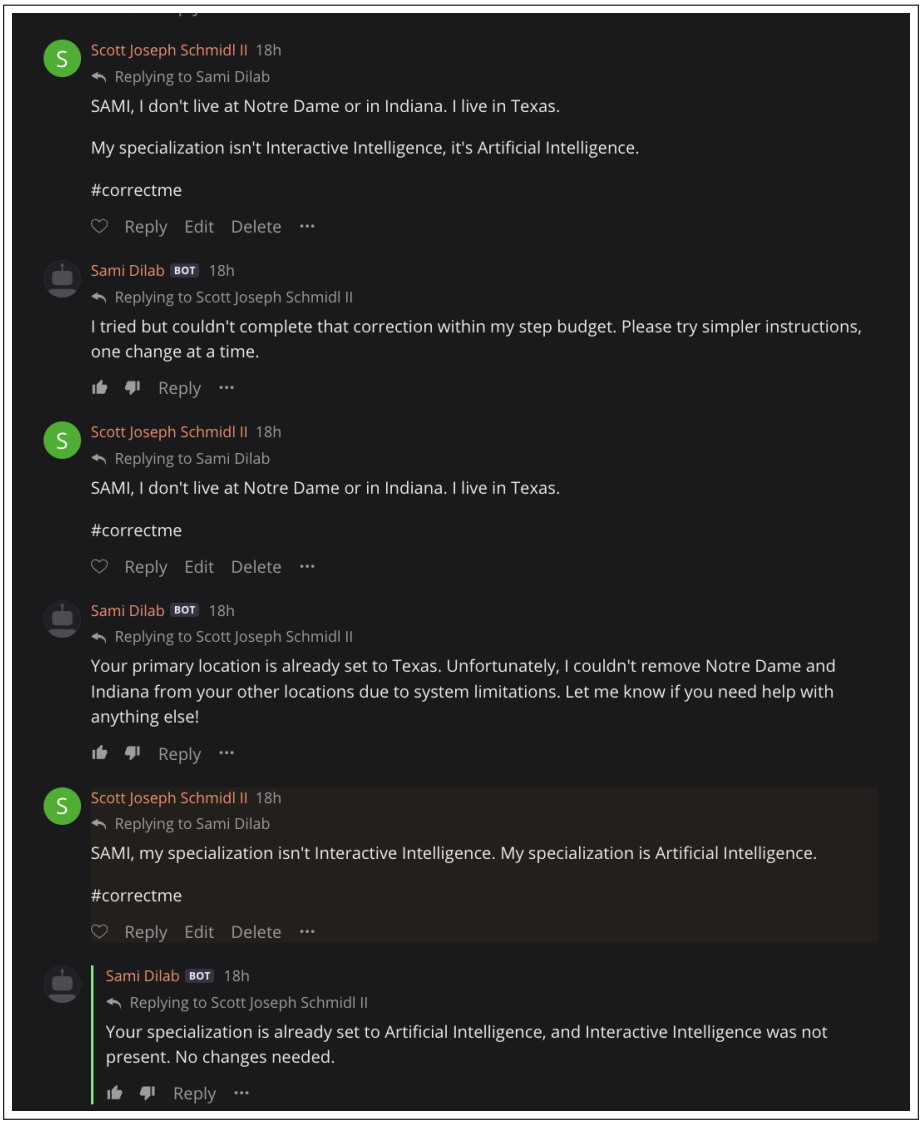


Figure 3—Correction to Location and Spec

1.5 Reflection on performance

From these interactions, I would characterize SAMI as a promising but flawed agent. On the positive side, it clearly built my profile, performed matchmaking, and understood parts of its architecture, but there are still things it gets wrong. There are clear errors in the knowledge graph and limitations to its capabilities, i.e. the location and specialization mistakes. There was potential for it to focus more on my experience with computer vision and my soon starting MBA at

Notre Dame when making matches. It seems that stating that I will be starting my MBA at Notre Dame drove it to automatically update its node about me with location information that was wrong; another flaw. The correction feedback loop is also limited. I wasn't able to provide multiple updates at once, but had to provide one at a time, which can become cumbersome.

These shortcomings point to three areas for improvement. The first is expanding and cleaning the knowledge graph, second, deepening its matchmaking ability, and strengthening the feedback loop so that corrections reliably update a students node in the graph.

2 THE REDESIGN CHALLENGE

2.1 Component selection

For this redesign, I'm prioritizing the "Interaction & Feedback Loop" component, because there were significant mistakes in the corrections I tried to give it. The concerning issues involved how SAMI made corrections and gave explanations over time. First, it said my specialization was Interactive Intelligence, then said it was Artificial Intelligence and that it never said Interactive Intelligence. Second, it said that it could not remove incorrect location information even though I don't live in Indiana and it acknowledged that I live in Texas. Third, it refused to update the graph when I gave it two changes at once. The problem undermines transparency and trust, because I can't reliably see nor shape how SAMI represents me, even though that representation is what drives matchmaking.

While the graph and matchmaking also have limitations, they are harder for students to fix with prompts. However, a stronger feedback loop would immediately improve my experience and make it easier for developers to debug the graph and matchmaking models in the future.

2.2 The critique

Currently, SAMI's interaction and feedback loop seems like a one-way FAQ instead of a conversational agent that can explain and update itself. When I ask for explanations, it mostly repeats my introduction and fixed description of itself instead of diving deeper into specific entities or they influenced my connections. When I ask it questions about me, it gives a detailed but somewhat inaccurate profile and when I try to correct them it struggles to make and communicate

those corrections.

The correction path had several failures that misrepresented me and hurt trust. Combined corrections are too much for it, so I had to find out on my own that I must send one correction at a time. Even then, SAMI only partially fixed its mistakes. It seems that the explanations and correction don't see each other, so I can't tell if it's profile about me matches what it tells, which reduces my confidence.

2.3 The high-level vision

By the end of the semester, I would expect SAMI's loop to feel like a trustworthy assistant instead of a static bot. When I ask what it knows about me, it should present a structured, accuracy profile grouped by topics. Furthermore, when I correct that information, it should apply the changes fully, update the graph, and confirm with me before and after changes.

Explanations should become personalized and tied to updates instead of repeating the generic spaCy/Neo4j information unless I ask for its architecture. SAMI should then explain matches using my current profile, i.e. "I matched you with this classmate, because you both live in Texas, are in the AI spec, and work with computer vision". A complete feedback loop would let students see a direct connection between their edits, its explanations, and the matches they receive.

2.4 The technical roadmap

1. Show Clear Profile. Add an explicit schema with the most important attributes of students, i.e. location, spec, and courses taken. Expose that information when #correctme is used as a structured summary instead of paragraph form. This will make it easier to understand what SAMI believes about me and make it easier to update corrections.
2. Make Changes Simple. When I send #correctme, SAMI should make explicit changes, such as, "set primary location to Texas. removed Indiana." or "change spec from II to AI", and then confirm each change individually. The reply should clearly state what worked and what didn't, saying something about "step budget" isn't helpful and gives no information about if any changes were actually made. This could have potentially been the issue with why the system gaslit me about my spec.
3. Base Explanations on Updates.Update #samiexplain so that it always uses the

current profile and match info when answering. For matches, SAMI shouldn't just list my interests as its reasoning for the matches but specifically enumerate what we have in common.